Question ID:101261
Topic Name: Mathematics-Section A
Question:

Let $R_1 = \{(a, b) \in \mathbf{N} \times \mathbf{N} : |a - b| \leq 13\}$ and
$R_2 = \{(a, b) \in \mathbf{N} \times \mathbf{N} : |a - b| \neq 13\}$. Then on $\mathbf{N}$:

A Both R_1 and R_2 are equivalence relations

B Neither R_1 nor R_2 is an equivalence relation

C R_1 is an equivalence relation but R_2 is not

D R_2 is an equivalence relation but R_1 is not

Answer Given By Candidate:**B**

Question ID:101262
Topic Name: Mathematics-Section A
Question:

Let $f(x)$ be a quadratic polynomial such that $f(-2) + f(3) = 0$. If one of the roots of $f(x) = 0$ is -1, then the sum of the roots of $f(x) = 0$ is equal to :

A $\frac{11}{3}$

B $\frac{7}{3}$

C $\frac{13}{3}$

D $\frac{14}{3}$

Answer Given By Candidate:**A**

Question ID:101263
Topic Name: Mathematics-Section A

Question:

The number of ways to distribute 30 identical candies among four children C_1, C_2, C_3 and C_4 so that C_2 receives atleast 4 and atmost 7 candies, C_3 receives atleast 2 and atmost 6 candies, is equal to :

A 205

B 615

C 510

D 430

Answer Given By Candidate:**B**

Question ID:101264
Topic Name: Mathematics-Section A
Question:

The term independent of x in the expansion of $(1 - x^2 + 3x^3)\left(\frac{5}{2}x^3 - \frac{1}{5x^2}\right)^{11}, x \neq 0$ is :

A $\frac{7}{40}$

B $\frac{33}{200}$

C $\frac{39}{200}$

D $\frac{11}{50}$

Answer Given By Candidate:**B**

Question ID:101265
Topic Name: Mathematics-Section A
Question:

If n arithmetic means are inserted between a and 100 such that the ratio of the first mean to the last mean is 1 : 7 and $a+n=33$, then the value of n is :

A 21

B 22

C 23

D 24

Answer Given By Candidate:**A**

Question ID:101266
Topic Name: Mathematics-Section A
Question:

Let $f, g : \mathbf{R} \to \mathbf{R}$ be functions defined by

$$f(x) = \begin{cases} [x] & , \quad x < 0 \\ |1 - x| & , \quad x \geq 0 \end{cases} \text{ and } g(x) = \begin{cases} e^x - x & , \quad x < 0 \\ (x-1)^2 - 1 & , \quad x \geq 0 \end{cases}$$

where $[x]$ denote the greatest integer less than or equal to x. Then, the function fog is discontinuous at exactly :

A one point

B two points

C three points

D four points

Answer Given By Candidate:**C**

Question ID:101267
Topic Name: Mathematics-Section A
Question:

Let $f : \mathbf{R} \to \mathbf{R}$ be a differentiable function such that $f\left(\frac{\pi}{4}\right) = \sqrt{2}$, $f\left(\frac{\pi}{2}\right) = 0$ and $f'\left(\frac{\pi}{2}\right) = 1$

and let $g(x) = \int_x^{\pi/4} \left(f'(t)\sec t + \tan t \sec f(t)\right) dt$ for $x \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right)$. Then $\lim\limits_{x \to \left(\frac{\pi}{2}\right)^-} g(x)$ is

equal to :

A 2

B 3

C 4

D -3

Answer Given By Candidate:**A**

Question ID:101268
Topic Name: Mathematics-Section A
Question:

Let $f : \mathbf{R} \to \mathbf{R}$ be a continuous function satisfying $f(x) + f(x+k) = n$, for all $x \in \mathbf{R}$ where $k > 0$

and n is a positive integer. If $I_1 = \int_0^{4nk} f(x)dx$ and $I_2 = \int_{-k}^{3k} f(x)dx$, then :

A $I_1 + 2I_2 = 4nk$

B $I_1 + 2I_2 = 2nk$

C $I_1 + nI_2 = 4n^2k$

D $I_1 + nI_2 = 6n^2k$

Answer Given By Candidate:**A**

Question ID:101269
Topic Name: Mathematics-Section A
Question:

The area of the bounded region enclosed by the curve $y = 3 - \left|x - \frac{1}{2}\right| - |x+1|$ and the x-axis is :

A $\frac{9}{4}$

B $\frac{45}{16}$

C $\frac{27}{8}$

D $\frac{63}{16}$

Answer Given By Candidate:**C**

Question ID:101270
Topic Name: Mathematics-Section A
Question:

Let $x = x(y)$ be the solution of the differential equation $2y\, e^{x/y^2}\, dx + \left(y^2 - 4xe^{x/y^2}\right)dy = 0$ such that $x(1) = 0$. Then, $x(e)$ is equal to :

A $e \log_e(2)$

B $-e \log_e(2)$

C $e^2 \log_e(2)$

D $-e^2 \log_e(2)$

Answer Given By Candidate:**B**

Question ID:101271
Topic Name: Mathematics-Section A
Question:

Let the slope of the tangent to a curve $y=f(x)$ at (x, y) be given by $2\tan x(\cos x-y)$. If the curve passes through the point $\left(\pi/4, 0\right)$, then the value of $\int_0^{\pi/2} y\,dx$ is equal to :

A
$(2-\sqrt{2})+\frac{\pi}{\sqrt{2}}$

B
$2-\frac{\pi}{\sqrt{2}}$

C
$(2+\sqrt{2})+\frac{\pi}{\sqrt{2}}$

D
$2+\frac{\pi}{\sqrt{2}}$

Answer Given By Candidate:**B**

Question ID:101272
Topic Name: Mathematics-Section A
Question:

Let a triangle be bounded by the lines $L_1: 2x+5y=10$; $L_2: -4x+3y=12$ and the line L_3, which passes through the point P(2, 3), intersects L_2 at A and L_1 at B. If the point P divides the line-segment AB, internally in the ratio 1 : 3, then the area of the triangle is equal to :

A
$\frac{110}{13}$

B
$\frac{132}{13}$

C
$\frac{142}{13}$

D
$\frac{151}{13}$

Answer Given By Candidate:**B**

Question ID:101273
Topic Name: Mathematics-Section A

Question:

Let $a > 0$, $b > 0$. Let e and l respectively be the eccentricity and length of the latus rectum of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Let e' and l' respectively be the eccentricity and length of the latus rectum of its conjugate hyperbola. If $e^2 = \frac{11}{14}l$ and $(e')^2 = \frac{11}{8}l'$, then the value of $77a + 44b$ is equal to :

A 100

B 110

C 120

D 130

Answer Given By Candidate:**C**

Question ID:101274
Topic Name: Mathematics-Section A
Question:

Let $\vec{a} = \alpha\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = -2\hat{i} + \alpha\hat{j} + \hat{k}$, where $\alpha \in \mathbf{R}$. If the area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $\sqrt{15(\alpha^2 + 4)}$, then the value of $2|\vec{a}|^2 + (\vec{a}\cdot\vec{b})|\vec{b}|^2$ is equal to :

A 10

B 7

C 9

D 14

Answer Given By Candidate:**B**

Question ID:101275
Topic Name: Mathematics-Section A
Question:

If vertex of a parabola is $(2, -1)$ and the equation of its directrix is $4x - 3y = 21$, then the length of its latus rectum is :

A 2

B 8

C 12

D 16

Answer Given By Candidate:B

Question ID:101276
Topic Name: Mathematics-Section A
Question:

Let the plane $ax+by+cz=d$ pass through $(2, 3, -5)$ and is perpendicular to the planes $2x+y-5z=10$ and

$3x+5y-7z=12$.

If a, b, c, d are integers $d > 0$ and gcd $(|a|, |b|, |c|, d)=1$, then the value of $a+7b+c+20d$ is equal to :

A 18

B 20

C 24

D 22

Answer Given By Candidate:D

Question ID:101277
Topic Name: Mathematics-Section A
Question:

The probability that a randomly chosen one-one function from the set {a, b, c, d} to the set {1, 2, 3, 4, 5} satisfies $f(a)+2f(b)-f(c)=f(d)$ is :

A $\frac{1}{24}$

B $\frac{1}{40}$

C $\frac{1}{30}$

D $\frac{1}{20}$

Answer Given By Candidate:D

Question ID:101278
Topic Name: Mathematics-Section A
Question:

The value of $\lim\limits_{n\to\infty} 6\tan\left\{\sum\limits_{r=1}^{n}\tan^{-1}\left(\frac{1}{r^2+3r+3}\right)\right\}$ is equal to :

A 1

B 2

C 3

D 6

Answer Given By Candidate:**D**

Question ID:101279
Topic Name: Mathematics-Section A
Question:

Let $\vec{a}$ be a vector which is perpendicular to the vector $3\hat{i} + \frac{1}{2}\hat{j} + 2\hat{k}$. If $\vec{a} \times \left(2\hat{i} + \hat{k}\right) = 2\hat{i} - 13\hat{j} - 4\hat{k}$, then the projection of the vector $\vec{a}$ on the vector $2\hat{i} + 2\hat{j} + \hat{k}$ is :

A $\frac{1}{3}$

B 1

C $\frac{5}{3}$

D $\frac{7}{3}$

Answer Given By Candidate:**C**

Question ID:101280
Topic Name: Mathematics-Section A
Question:

If $\cot\alpha = 1$ and $\sec\beta = -\frac{5}{3}$, where $\pi < \alpha < \frac{3\pi}{2}$ and $\frac{\pi}{2} < \beta < \pi$, then the value of $\tan(\alpha+\beta)$ and the quadrant in which $\alpha+\beta$ lies, respectively are :

A $-\frac{1}{7}$ and IVth quadrant

B 7 and Ist quadrant

C -7 and IVth quadrant

D

$\frac{1}{7}$ and Ist quadrant

Answer Given By Candidate:**A**

Question ID:101281
Topic Name: Mathematics-Section B
Question:

Let the image of the point P(1, 2, 3) in the line $L: \frac{x-6}{3} = \frac{y-1}{2} = \frac{z-2}{3}$ be Q. Let R (α, β, γ) be a point that divides internally the line segment PQ in the ratio 1 : 3. Then the value of $22(\alpha+\beta+\gamma)$ is equal to _____.

Answer Given By Candidate:**768**

Question ID:101282
Topic Name: Mathematics-Section B
Question:

Suppose a class has 7 students. The average marks of these students in the mathematics examination is 62, and their variance is 20. A student fails in the examination if he/she gets less than 50 marks, then in worst case, the number of students can fail is _____.

Answer Given By Candidate:**4**

Question ID:101283
Topic Name: Mathematics-Section B
Question:

If one of the diameters of the circle $x^2+y^2-2\sqrt{2}x - 6\sqrt{2}y + 14 = 0$ is a chord of the circle $(x - 2\sqrt{2})^2 + (y - 2\sqrt{2})^2 = r^2$, then the value of r^2 is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101284
Topic Name: Mathematics-Section B
Question:

If $\lim\limits_{x \to 1} \frac{\sin(3x^2 - 4x + 1) - x^2 + 1}{2x^3 - 7x^2 + ax + b} = -2$, then the value of (a − b) is equal to _____ .

Answer Given By Candidate:**11**

Question ID:101285
Topic Name: Mathematics-Section B

Question:

Let for $n=1, 2, \ldots, 50$, S_n be the sum of the infinite geometric progression whose first term is n^2 and whose common ratio is $\frac{1}{(n+1)^2}$. Then the value of $\frac{1}{26}+\sum_{n=1}^{50}\left(S_n+\frac{2}{n+1}-n-1\right)$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101286
Topic Name: Mathematics-Section B
Question:

If the system of linear equations

$2x-3y=\gamma+5,$

$\alpha x+5y=\beta+1$, where $\alpha, \beta, \gamma \in \mathbf{R}$ has infinitely many solutions, then the value of $|9\alpha+3\beta+5\gamma|$ is equal to _____.

Answer Given By Candidate:**58**

Question ID:101287
Topic Name: Mathematics-Section B
Question:

Let $A=\begin{pmatrix}1+i & 1\\ -i & 0\end{pmatrix}$ where $i=\sqrt{-1}$. Then, the number of elements in the set $\{n \in \{1, 2, \ldots, 100\} : A^n = A\}$ is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101288
Topic Name: Mathematics-Section B
Question:

Sum of squares of modulus of all the complex numbers z satisfying $\bar{z}=iz^2+z^2-z$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101289
Topic Name: Mathematics-Section B
Question:

Let $S=\{1, 2, 3, 4\}$. Then the number of elements in the set $\{f : S\times S \to S : f$ is onto and $f(a, b)=f(b, a) \geq a \ \forall\ (a, b) \in S\times S\}$ is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:101290

Topic Name: Mathematics-Section B
Question:

The maximum number of compound propositions, out of $p \vee r \vee s$, $p \vee r \vee \sim s$, $p \vee \sim q \vee s$, $\sim p \vee \sim r \vee s$, $\sim p \vee \sim r \vee \sim s$, $\sim p \vee q \vee \sim s$, $q \vee r \vee \sim s$, $q \vee \sim r \vee \sim s$, $\sim p \vee \sim q \vee \sim s$ that can be made simultaneously true by an assignment of the truth values to p, q, r and s, is equal to _____.

Answer Given By Candidate:7

Question ID:101201
Topic Name: Physics-Section A
Question:

Velocity (v) and acceleration (a) in two systems of units 1 and 2 are related as $v_2 = \frac{n}{m^2} v_1$ and $a_2 = \frac{a_1}{mn}$ respectively. Here m and n are constants. The relations for distance and time in two systems respectively are :

A
$\frac{n^3}{m^3} L_1 = L_2$ and $\frac{n^2}{m} T_1 = T_2$

B
$L_1 = \frac{n^4}{m^2} L_2$ and $T_1 = \frac{n^2}{m} T_2$

C
$L_1 = \frac{n^2}{m} L_2$ and $T_1 = \frac{n^4}{m^2} T_2$

D
$\frac{n^2}{m} L_1 = L_2$ and $\frac{n^4}{m^2} T_1 = T_2$

Answer Given By Candidate:B

Question ID:101202
Topic Name: Physics-Section A
Question:

A ball is spun with angular acceleration $\alpha = 6t^2 - 2t$ where t is in second and α is in $rads^{-2}$. At $t = 0$, the ball has angular velocity of 10 $rads^{-1}$ and angular position of 4 rad. The most appropriate expression for the angular position of the ball is :

A
$\frac{3}{2} t^4 - t^2 + 10t$

B
$\frac{t^4}{2} - \frac{t^3}{3} + 10t + 4$

C

$\frac{2t^4}{3} - \frac{t^3}{6} + 10t + 12$

D

$2t^4 - \frac{t^3}{2} + 5t + 4$

Answer Given By Candidate:**B**

Question ID:101203
Topic Name: Physics-Section A
Question:

A block of mass 2 kg moving on a horizontal surface with speed of 4 ms^{-1} enters a rough surface ranging from $x = 0.5$ m to $x = 1.5$ m. The retarding force in this range of rough surface is related to distance by $F = -kx$ where $k = 12\ Nm^{-1}$. The speed of the block as it just crosses the rough surface will be :

A zero

B 1.5 ms^{-1}

C 2.0 ms^{-1}

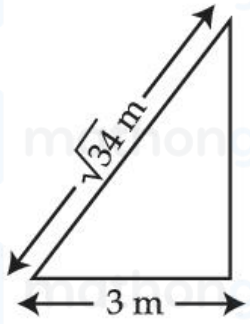
D 2.5 ms^{-1}

Answer Given By Candidate:**C**

Question ID:101204
Topic Name: Physics-Section A
Question:

A $\sqrt{34}$ m long ladder weighing 10 kg leans on a frictionless wall. Its feet rest on the floor 3 m away from the wall as shown in the figure. If F_f and F_w are the reaction forces of the floor and the wall, then ratio of F_w/F_f will be :

(Use $g = 10\ m/s^2$.)



A $\frac{6}{\sqrt{110}}$

B

$\frac{3}{\sqrt{113}}$

C $\frac{3}{\sqrt{109}}$

D $\frac{2}{\sqrt{109}}$

Answer Given By Candidate:**C**

Question ID:101205
Topic Name: Physics-Section A
Question:

Water falls from a 40 m high dam at the rate of 9×10^4 kg per hour. Fifty percentage of gravitational potential energy can be converted into electrical energy. Using this hydro electric energy number of 100 W lamps, that can be lit, is :

(Take $g=10\ ms^{-2}$)

A 25

B 50

C 100

D 18

Answer Given By Candidate:**C**

Question ID:101206
Topic Name: Physics-Section A
Question:

Two objects of equal masses placed at certain distance from each other attracts each other with a force of F. If one-third mass of one object is transferred to the other object, then the new force will be :

A $\frac{2}{9}$ F

B $\frac{16}{9}$ F

C $\frac{8}{9}$ F

D F

Answer Given By Candidate:**C**

Question ID:101207
Topic Name: Physics-Section A
Question:

A water drop of radius 1 μm falls in a situation where the effect of buoyant force is negligible. Co-efficient of viscosity of air is $1.8\times10^{-5}\ \text{Nsm}^{-2}$ and its density is negligible as compared to that of water $10^6\ \text{gm}^{-3}$. Terminal velocity of the water drop is :

(Take acceleration due to gravity $=10\ \text{ms}^{-2}$)

A $145.4\times10^{-6}\ \text{ms}^{-1}$

B $118.0\times10^{-6}\ \text{ms}^{-1}$

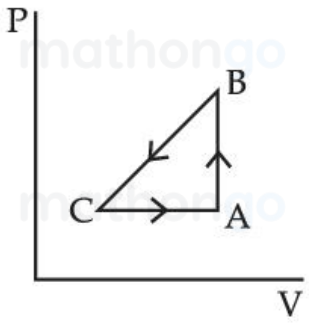
C $132.6\times10^{-6}\ \text{ms}^{-1}$

D $123.4\times10^{-6}\ \text{ms}^{-1}$

Answer Given By Candidate:**D**

Question ID:101208
Topic Name: Physics-Section A
Question:

A sample of an ideal gas is taken through the cyclic process ABCA as shown in figure. It absorbs, 40 J of heat during the part AB, no heat during BC and rejects 60 J of heat during CA. A work of 50 J is done on the gas during the part BC. The internal energy of the gas at A is 1560 J. The workdone by the gas during the part CA is :



A 20 J

B 30 J

C −30 J

D −60 J

Answer Given By Candidate:**B**

Question ID:101209
Topic Name: Physics-Section A
Question:

What will be the effect on the root mean square velocity of oxygen molecules if the temperature is doubled and oxygen molecule dissociates into atomic oxygen ?

A The velocity of atomic oxygen remains same

B The velocity of atomic oxygen doubles

C The velocity of atomic oxygen becomes half

D The velocity of atomic oxygen becomes four times

Answer Given By Candidate:**B**

Question ID:101210
Topic Name: Physics-Section A
Question:

Two point charges A and B of magnitude $+8\times10^{-6}$ C and -8×10^{-6} C respectively are placed at a distance d apart. The electric field at the middle point O between the charges is 6.4×10^{4} NC^{-1}. The distance 'd' between the point charges A and B is :

A 2.0 m

B 3.0 m

C 1.0 m

D 4.0 m

Answer Given By Candidate:**B**

Question ID:101211
Topic Name: Physics-Section A
Question:

Resistance of the wire is measured as 2 Ω and 3 Ω at 10°C and 30°C respectively. Temperature co-efficient of resistance of the material of the wire is :

A $0.033\ °C^{-1}$

B $-0.033\ °C^{-1}$

C $0.011\ °C^{-1}$

D $0.055\ °C^{-1}$

Answer Given By Candidate:**A**

Question ID:101212
Topic Name: Physics-Section A
Question:

The space inside a straight current carrying solenoid is filled with a magnetic material having magnetic susceptibility equal to 1.2×10^{-5}. What is fractional increase in the magnetic field inside solenoid with respect to air as medium inside the solenoid ?

A 1.2×10^{-5}

B 1.2×10^{-3}

C 1.8×10^{-3}

D 2.4×10^{-5}

Answer Given By Candidate:**B**

Question ID:101213
Topic Name: Physics-Section A
Question:

Two parallel, long wires are kept 0.20 m apart in vacuum, each carrying current of x A in the same direction. If the force of attraction per meter of each wire is 2×10^{-6} N, then the value of x is approximately :

A 1

B 2.4

C 1.4

D 2

Answer Given By Candidate:**C**

Question ID:101214
Topic Name: Physics-Section A
Question:

A coil is placed in a time varying magnetic field. If the number of turns in the coil were to be halved and the radius of wire doubled, the electrical power dissipated due to the current induced in the coil would be :

(Assume the coil to be short circuited.)

A Halved

B Quadrupled

C The same

D Doubled

Answer Given By Candidate:**D**

Question ID:101215
Topic Name: Physics-Section A
Question:

An EM wave propagating in x-direction has a wavelength of 8 mm. The electric field vibrating y-direction has maximum magnitude of 60 Vm^{-1}. Choose the correct equations for electric and magnetic fields if the EM wave is propagating in vacuum :

A

$$E_y = 60\sin\left[\frac{\pi}{4} \times 10^3 \ (x - 3 \times 10^8 t)\right]\hat{j} \ \text{Vm}^{-1}$$

$$B_z = 2\sin\left[\frac{\pi}{4} \times 10^3 \ (x - 3 \times 10^8 t)\right]\hat{k} \ \text{T}$$

B

$$E_y = 60\sin\left[\frac{\pi}{4} \times 10^3 \ (x - 3 \times 10^8 t)\right]\hat{j} \ \text{Vm}^{-1}$$

$$B_z = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^3 \ (x - 3 \times 10^8 t)\right]\hat{k} \ \text{T}$$

C

$$E_y = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^3 \ (x - 3 \times 10^8 t)\right]\hat{j} \ \text{Vm}^{-1}$$

$$B_z = 60\sin\left[\frac{\pi}{4} \times 10^3 \ (x - 3 \times 10^8 t)\right]\hat{k} \ \text{T}$$

D

$$E_y = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^4 \ (x - 4 \times 10^8 t)\right]\hat{j} \ \text{Vm}^{-1}$$

$$B_z = 60\sin\left[\frac{\pi}{4} \times 10^4 \ (x - 4 \times 10^8 t)\right]\hat{k} \ \text{T}$$

Answer Given By Candidate:**B**

Question ID:101216
Topic Name: Physics-Section A
Question:

In young's double slit experiment performed using a monochromatic light of wavelength λ, when a glass plate ($\mu = 1.5$) of thickness $x\lambda$ is introduced in the path of the one of the interfering beams, the intensity at the position where the central maximum occurred previously remains unchanged. The value of x will be :

A 3

B 2

C 1.5

D 0.5

Answer Given By Candidate:**B**

Question ID:101217
Topic Name: Physics-Section A

Question:

Let K_1 and K_2 be the maximum kinetic energies of photo-electrons emitted when two monochromatic beams of wavelength λ_1 and λ_2, respectively are incident on a metallic surface. If $\lambda_1 = 3\lambda_2$ then :

A

$$K_1 > \frac{K_2}{3}$$

B

$$K_1 < \frac{K_2}{3}$$

C

$$K_1 = \frac{K_2}{3}$$

D

$$K_2 = \frac{K_1}{3}$$

Answer Given By Candidate:**B**

Question ID:101218
Topic Name: Physics-Section A
Question: